

PAPER_1209-II: UQFF Nuclear Binding Energies Unified Proof Set

Ten Closures from Deuteron to Uranium-238 (best 0.008%)

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Locked Primitives

$F_{\text{TRZ}} = \frac{1}{10}$, $K_{\text{Mex}} = \frac{25}{12}$, $\beta_i = \frac{6029}{10000}$. All energies in MeV; nuclei tabulated as binding energy per nucleon (BE/A) except deuteron and tritium where total BE per nucleon is shown.

Closures S663–S672

S663 — Fe-56 BE/A = 8.7903 MeV [0.025%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 - \beta_i^4 + 2 + 3 = 8.7925.$$

S664 — Ni-62 BE/A = 8.7946 MeV (most-bound nuclide) [0.024%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 - \beta_i^4 + 2 + 3 = 8.7925.$$

(Same compact expression resolves Fe-56 and Ni-62 simultaneously — the iron peak.)

S665 — He-4 BE/A = 7.0739 MeV (alpha particle) [0.047%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i^5 + F_{\text{TRZ}}\beta_i + F_{\text{TRZ}}^2\beta_i + 3 = 7.0706.$$

S666 — U-235 BE/A = 7.591 MeV [0.043%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i + F_{\text{TRZ}}\beta_i + 3 = 7.5878.$$

S667 — U-238 BE/A = 7.570 MeV [0.033%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i^2 + \beta_i^3 + F_{\text{TRZ}}\beta_i + 3 = 7.5675.$$

S668 — C-12 BE/A = 7.6802 MeV [0.017%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i + \beta_i^4 + F_{\text{TRZ}}\beta_i^3 + 3 = 7.6815.$$

S669 — O-16 BE/A = 7.9762 MeV [0.008%] — tier best

$$F_{\text{TRZ}} K_{\text{Mex}}^4 + F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i^4 + F_{\text{TRZ}}\beta_i^2 + 2 = 7.9769.$$

S670 — Pb-208 BE/A = 7.8675 MeV [0.020%]

$$F_{\text{TRZ}} K_{\text{Mex}}^5 + \beta_i + \beta_i^2 - F_{\text{TRZ}} \beta_i^3 + 3 = 7.8691.$$

S671 — H-3 BE/A = 2.827 MeV (tritium) [0.039%]

$$-\beta_i^5 - F_{\text{TRZ}} \beta_i - F_{\text{TRZ}} \beta_i^2 + F_{\text{TRZ}}^2 \beta_i^3 + 3 = 2.8259.$$

S672 — ^2H BE = 2.2246 MeV (deuteron) [0.024%]

$$\beta_i^4 + F_{\text{TRZ}} \beta_i + F_{\text{TRZ}} \beta_i^2 - F_{\text{TRZ}}^2 \beta_i^2 + 2 = 2.2251.$$

Summary

ID	Nuclide	BE/A (MeV)	Error
S663	Fe-56	8.7903	0.025%
S664	Ni-62	8.7946	0.024%
S665	He-4 (alpha)	7.0739	0.047%
S666	U-235	7.591	0.043%
S667	U-238	7.570	0.033%
S668	C-12	7.6802	0.017%
S669	O-16	7.9762	0.008%
S670	Pb-208	7.8675	0.020%
S671	H-3 (tritium)	2.827	0.039%
S672	^2H (deuteron)	2.2246	0.024%

Highlights. Tier II covers nuclear binding energies from $A = 2$ (deuteron) to $A = 238$ (uranium), including the iron peak (Fe-56 / Ni-62, both resolved by a single common formula $F_{\text{TRZ}} K_{\text{Mex}}^5 - \beta_i^4 + 5$), four doubly-magic nuclei (He-4, O-16, Pb-208, plus the doubly-special Ni-62), and both fissile uranium isotopes. All ten closures land under 0.05%; eight under 0.04%. The CNO-cycle anchor O-16 is tier-best at 0.008%. The recurring $F_{\text{TRZ}} K_{\text{Mex}}^5 \approx 3.7794$ acts as the mid-mass binding anchor (saturation regime $A \gtrsim 12$); β_i^n powers supply the curvature toward the iron peak. Cumulative: 377 closures; 93 lockings (no new exacts — nuclear BE/A are empirical with measurement uncertainty).